THANMAYEE BOYAPATI

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EDUCATION

University of Minnesota -Twin Cities December 2025

Major: Computer Science B.S & Data Science B.S | **Minor:** Statistics

SKILLS

Languages: Python, SQL, Java, JavaScript, C, R,

ML/AI: PyTorch, Scikit-learn, Hugging Face, Sentence Transformers

GenAI: LLM Fine-Tuning, RAG, Embeddings, Prompt Engineering, Evaluation Metrics

Data & Systems: Pandas, NumPy, Docker, AWS, GCP, Linux, Jupyter, MongoDB, REST APIs, Tableau

RELEVANT EXPERIENCE

Mayo Clinic | Rochester, MN June 2025 - December 2025

AI Engineer Intern - BioStatistician Department

- Built an end-to-end genomics question-answering system by integrating OCR, entity extraction, embedding-based retrieval, and LLM-based generation, enabling accurate and grounded responses to domain-specific clinical queries
- Architected a retrieval layer using MongoDB and LangChain to index, store, and query extracted clinical entities and document chunks, enabling dynamic access to structured and unstructured data during LLM inference
- Fine-tuned and evaluated a domain-adapted LLaMA-4 model using embedding-based retrieval and Sentence Transformer similarity metrics, iterating on retrieval strategies and context selection to improve contextual relevance

Group Lens Research Lab | Minneapolis, MN August 2024 - Present

Undergraduate Researcher

- Implemented a hybrid intent-alignment scoring pipeline combining NLI models (RoBERTa) and semantic embeddings (Sentence Transformers) to quantify alignment between user intent and news headlines for personalization assessment
- Created diagnostic analytics and visualization workflows to surface click-through-rate trends and locality-category shifts, enabling rapid identification of failure modes across headline generation strategies
- Designed task-specific evaluation metrics and executed controlled experiments across prompt and topic variations to analyze model behavior, supporting a forthcoming publication on news personalization

PROJECTS

Finance Tracker

- Designed and implemented a full-stack personal finance platform using React, Node.js, Express, and MongoDB to structure, and retrieve transaction records, enabling flexible categorization, filtering, and analytics-ready persistence
- Integrated automated expense categorization and LLM-driven narrative summaries to convert unstructured financial activity into clear, personalized spending insights that support informed budgeting and decision-making

Brain-Based Prediction of tDCS Treatment Response

- Compared multiple brain connectivity representations to predict individualized treatment response through functional connectivity features from clinical neuroimaging data, evaluating representational tradeoffs in predictive performance
- Applied machine learning approaches, like T-Learner, X-Learner, &Causal Forests, with dimensionality reduction to estimate personalized treatment effects while balancing expressiveness and generalization under small-sample conditions

VPN-Based Remote Robotics Interface

- Engineered and launched a Site-to-Site VPN to provide secure communication between autonomous robots and remote driver stations, integrating it with robotics software for reliable real-time operation during testing and competitions
- Facilitated collaboration during the pandemic by providing VPN configuration support to multiple companies and global teams, including those based in Poland, enabling secure access to network resources

HONORS & LEADERSHIP

- NCWIT Aspirations in Computing – *State Winner*
- College of Science and Engineering – *Dean's List*
- CSCI 5525 Machine Learning – *Teaching Assistant*
- South Indian Student Association (SISA) – *Co-Founder*
- NCWIT – National Center for Women & IT – *Campus Representative*
- College of Science & Engineering – *CSE Ambassador*
- Northside STEM Bridge Program – *STEM Facilitator*

INTERESTS

Dance | Cooking | Escape Rooms | Volunteering | Roller Coasters | Books | Horse Riding